
Inventory Control System

DBMS Mini-Project | 3rd Year B.Tech

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The Problem

- ⚠️ **Manual Tracking:** Paper logs cause critical data loss.
- 📦 **Stockouts:** Unnoticed low inventory disrupts supply chains.
- 🚫 **No Visibility:** Lack of real-time inventory valuation.



System Objectives



Automation

Eliminate manual stock calculations via triggers.



Insights

Provide real-time dashboard analytics.



Security

Enforce role-based multi-user authentication.

Technology Stack



Python

Core application logic.



SQLite 3

Relational database engine.



Streamlit

Frontend web framework.



Pandas

Data manipulation and routing.

Core Entity-Relationship Model

Table: Products

PK product_id (Integer)
product_name (Text)
unit_price (Real)
quantity_in_stock (Integer)
reorder_level (Integer)



Table: Transactions

PK transaction_id (Text)
FK product_id (Integer)
transaction_type (Text)
quantity (Integer)
transaction_date (Timestamp)

Automated Database Triggers

Stock levels update instantly at the database layer. No manual logic required.

```
CREATE TRIGGER after_transaction_insert_in
AFTER INSERT ON Transactions
WHEN NEW.transaction_type = 'IN'
BEGIN
    UPDATE Products
    SET quantity_in_stock = quantity_in_stock + NEW.quantity
    WHERE product_id = NEW.product_id;
END;
```

System Features

Full CRUD Capabilities

Add, Edit, and Delete products seamlessly.

Alphanumeric IDs

Generated tracking codes like TXN-A8F3K9M2L1.

Data Integrity

Deleting a product removes history via foreign key logic.

Multi-User Login

Secure access control for authorized admins.

Real-Time Operations

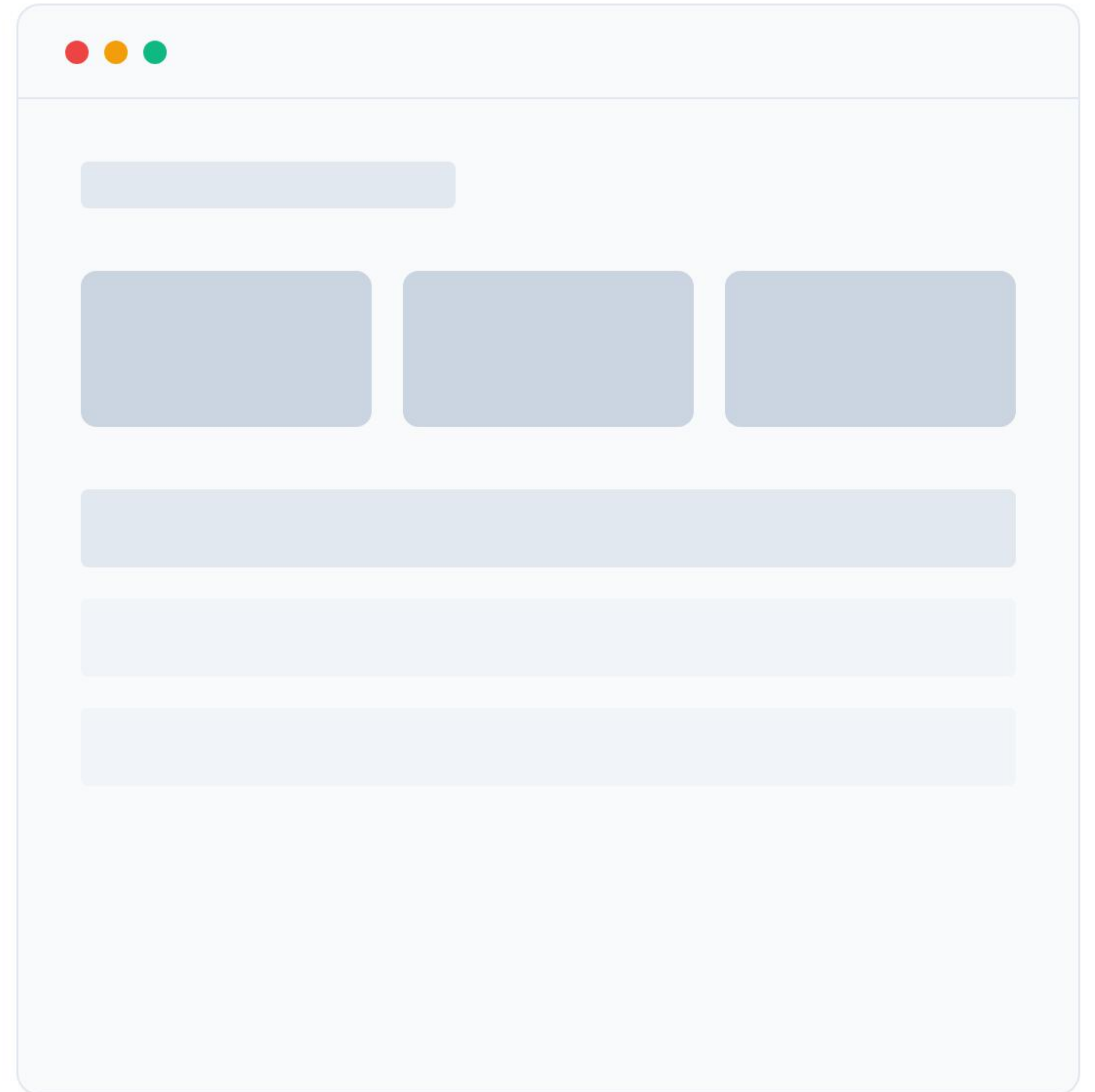
Transactions are reflected instantly across the unified dashboard.

100%

Accuracy in stock reconciliation.

Zero

Manual data entry required.



Thank You

Questions?

Abhay Kumar & Yash Pandey